

A Finite-Dimensional Counterexample to Tensoring Centers for Banach Algebras

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Status

This packet records a finite-dimensional counterexample to the unrestricted Banach-algebra analogue of the center theorem proved by Gupta and Jain for C*-algebras. The source paper proves that, for C*-algebras A and B ,

$$\mathcal{Z}(A \otimes_{\gamma} B) = \mathcal{Z}(A) \otimes_{\gamma} \mathcal{Z}(B),$$

and then asks whether there is a similar relationship for Banach algebras.

The example below shows that the direct equality fails for Banach algebras, even in dimension $4 \cdot 2$.

Source Question

After proving the C*-algebra identity

$$\mathcal{Z}(A \otimes_{\gamma} B) = \mathcal{Z}(A) \otimes_{\gamma} \mathcal{Z}(B),$$

Gupta and Jain ask:

Is there any relationship, as above, between the center of $A \otimes_{\gamma} B$ and $\mathcal{Z}(A) \otimes_{\gamma} \mathcal{Z}(B)$ for Banach algebras A and B ?

A General Annihilator Mechanism

For a Banach algebra B , write

$$\text{Ann}(B) = \{b \in B : bB = Bb = \{0\}\}.$$

Lemma 1. *Let A and B be Banach algebras. If $0 \neq b \in \text{Ann}(B)$, then*

$$A \otimes \mathbb{C}b \subseteq \mathcal{Z}(A \otimes_{\gamma} B).$$

Consequently, if A is noncommutative and $b \neq 0$, then the center of $A \otimes_{\gamma} B$ may contain elements not coming from $\mathcal{Z}(A) \otimes_{\gamma} \mathcal{Z}(B)$.

Proof. For elementary tensors,

$$(a \otimes b)(x \otimes y) = ax \otimes by = 0, \quad (x \otimes y)(a \otimes b) = xa \otimes yb = 0$$

for every $a, x \in A$ and $y \in B$. Hence every $a \otimes b$ commutes with every elementary tensor. Since multiplication on $A \otimes_{\gamma} B$ is continuous and the algebraic tensor product is dense, $a \otimes b$ is central in $A \otimes_{\gamma} B$. $\square$

The Counterexample

Theorem 1. *There are finite-dimensional Banach algebras A and B such that*

$$\mathcal{Z}(A) \otimes_\gamma \mathcal{Z}(B) \subsetneq \mathcal{Z}(A \otimes_\gamma B).$$

Proof. Let $A = M_2(\mathbb{C})$ with its usual operator norm. Let $B = \mathbb{C}e \oplus \mathbb{C}n$ with multiplication

$$e^2 = e, \quad en = ne = n^2 = 0,$$

and norm $\|\alpha e + \beta n\| = |\alpha| + |\beta|$. This is a finite-dimensional Banach algebra. It is commutative, so $\mathcal{Z}(B) = B$, and

$$\mathcal{Z}(A) = \mathbb{C}I.$$

Because all spaces are finite-dimensional, the projective tensor product is the algebraic tensor product equipped with a complete norm. Every element of $A \otimes_\gamma B$ has a unique form

$$X \otimes e + Y \otimes n, \quad X, Y \in A.$$

Let $T, S \in A$. The multiplication rules in B give

$$(X \otimes e + Y \otimes n)(T \otimes e + S \otimes n) = XT \otimes e$$

and

$$(T \otimes e + S \otimes n)(X \otimes e + Y \otimes n) = TX \otimes e.$$

Therefore $X \otimes e + Y \otimes n$ is central in $A \otimes_\gamma B$ if and only if

$$XT = TX \quad \text{for every } T \in M_2(\mathbb{C}).$$

This is equivalent to $X \in \mathbb{C}I$, while $Y \in M_2(\mathbb{C})$ is arbitrary. Hence

$$\mathcal{Z}(A \otimes_\gamma B) = \mathbb{C}I \otimes e + M_2(\mathbb{C}) \otimes n.$$

On the other hand,

$$\mathcal{Z}(A) \otimes_\gamma \mathcal{Z}(B) = \mathbb{C}I \otimes_\gamma B = \mathbb{C}I \otimes e + \mathbb{C}I \otimes n.$$

The inclusion is strict. For instance, if E_{12} is the standard matrix unit, then $E_{12} \otimes n \in \mathcal{Z}(A \otimes_\gamma B)$. It cannot belong to $\mathcal{Z}(A) \otimes_\gamma \mathcal{Z}(B)$: applying the coordinate functional e^* on B forces the $I \otimes e$ coefficient to be zero, and applying n^* would then force $E_{12} \in \mathbb{C}I$, which is impossible. $\square$

What This Does and Does Not Settle

The example gives a negative answer to the direct unrestricted Banach-algebra extension of the C^* -algebra identity. The obstruction is not analytic subtlety of the projective norm; it is algebraic annihilator mass in a nonunital Banach algebra.

It does not address stronger formulations in which both Banach algebras are unital, or more generally have bounded approximate identities. In those cases a nonzero two-sided annihilator is impossible, so this particular obstruction is removed.

Novelty Check

The local run indexes were searched for arXiv:1809.01131, the source title, and keywords around centers of projective tensor products of Banach algebras. No prior packet for this question was present. External searches for the center question and for the annihilator mechanism found the source paper and adjacent C^* -algebra tensor-product literature, but no direct recorded use of this finite-dimensional example as an answer to the final Banach-algebra question.

References

- [1] V. P. Gupta and R. Jain, *On Banach space projective tensor product of C^* -algebras*, arXiv:1809.01131, 2018.